

TOLEDO EXECUTIVE, OH, USA

WMO: 724287

Lat: 41.563N

Lon: 83.476W

Elev: 622

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 04848

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.2	6.6	-6.8	3.9	2.7	-1.2	5.3	8.2	28.0	28.1	26.0	28.2	8.4	230	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	91.3	73.9	88.8	72.7	86.1	71.4	77.1	87.1	75.4	84.0	73.9	82.2	11.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	129.0	82.3	72.5	123.4	80.6	71.5	119.1	79.3	40.8	87.0	39.2	84.1	37.8	82.1	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.9	21.7	19.2	DB	-4.5	95.2	5.8	3.7	-8.7	97.8	-12.1	100.0	-15.4	102.0	-19.6	104.7	(4)
			WB	-4.7	79.3	5.6	1.5	-8.7	80.4	-12.0	81.2	-15.1	82.1	-19.2	83.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	26.1	28.6	37.7	49.8	61.5	71.0	74.7	72.6	66.2	54.0	42.4	31.3
	DBStd	19.29	11.43	10.94	10.64	8.88	8.55	6.60	5.55	4.90	7.21	8.72	9.35	10.30
	HDD50	2772	742	603	405	110	8	0	0	0	1	54	264	585
	HDD65	5885	1205	1019	846	462	173	22	1	4	70	359	679	1045
	CDD50	3298	2	4	25	103	365	629	765	701	486	178	35	5
	CDD65	938	0	0	1	6	65	201	302	241	105	17	0	0
	CDH74	7957	0	0	11	85	618	1722	2730	1826	829	134	2	0
	CDH80	2663	0	0	1	12	189	617	1028	528	259	29	0	0
Wind	WSAvg	8.9	10.8	10.4	9.8	10.5	8.9	8.1	7.1	6.4	6.9	8.7	9.6	10.0
Precipitation	PrecAvg	33.7	2.0	2.0	2.6	3.2	3.3	3.6	3.2	3.1	2.9	2.3	2.7	2.6
	PrecMax	47.8	4.6	5.7	5.7	6.3	6.8	8.5	9.2	8.5	8.1	6.3	7.2	6.8
	PrecMin	22.0	0.3	0.3	0.6	0.9	0.9	0.3	0.4	0.4	0.6	0.3	0.6	0.6
	PrecStd	5.2	1.1	1.3	1.2	1.3	1.5	1.7	1.7	1.8	1.6	1.3	1.5	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.5	64.7	74.1	81.8	89.7	93.8	95.6	91.4	91.1	83.9	72.3	62.6
		MCWB	54.6	55.7	62.8	62.8	70.7	73.7	76.2	74.6	74.0	69.5	59.4	58.1
	2%	DB	52.3	55.4	65.6	75.7	85.1	90.3	91.9	88.3	86.9	77.5	64.6	55.7
		MCWB	49.3	51.7	55.9	60.5	68.8	72.5	74.5	74.0	71.4	64.8	56.6	52.5
	5%	DB	46.4	48.3	60.4	71.0	81.4	87.2	89.4	85.4	82.3	72.7	61.3	50.3
		MCWB	43.3	43.5	53.2	58.2	67.0	71.2	73.5	72.1	69.1	62.8	54.4	46.5
	10%	DB	40.8	42.7	54.7	65.6	76.4	83.3	85.9	82.5	78.9	68.3	56.9	44.9
		MCWB	37.8	38.6	48.7	55.5	64.7	69.1	71.8	70.7	67.3	60.2	51.1	41.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.0	58.1	63.9	66.3	73.7	77.1	79.4	78.3	76.6	71.5	62.2	58.8
		MCDB	57.8	62.7	73.4	76.7	85.2	89.7	91.1	87.8	87.7	81.8	67.5	62.5
	2%	WB	49.9	51.3	58.2	62.9	71.0	74.9	77.5	76.7	73.9	67.9	58.4	53.1
		MCDB	52.0	54.8	64.3	72.4	81.4	85.9	88.3	84.5	82.2	74.1	63.6	55.4
	5%	WB	43.9	44.1	54.2	59.7	68.7	73.2	75.7	74.7	71.7	64.5	55.0	46.5
		MCDB	46.6	47.6	59.3	69.0	78.1	83.4	84.4	81.6	79.5	70.7	60.5	49.4
	10%	WB	38.0	38.5	49.0	56.5	66.4	71.4	74.1	73.0	69.6	61.1	51.5	41.9
		MCDB	40.4	42.3	54.6	64.7	74.6	80.2	82.3	79.8	75.9	67.5	56.6	44.7
Mean Daily Temperature Range	5% DB	MDBR	13.4	14.2	16.5	19.2	19.1	18.4	18.4	17.7	19.1	17.8	15.2	12.2
		MCDBR	17.8	21.7	25.0	27.1	24.2	23.0	21.4	20.7	23.8	23.4	21.6	17.2
		MCWBR	15.6	17.5	17.2	15.4	12.1	10.0	8.6	9.3	11.0	12.9	15.8	15.0
	5% WB	MCDBR	17.3	20.8	23.4	24.5	21.4	20.0	18.6	17.6	19.1	20.1	19.4	16.6
		MCWBR	15.8	17.7	17.6	16.1	11.8	10.1	8.4	9.2	10.1	12.2	15.7	15.4
Clear-Sky Solar Irradiance	taub	0.302	0.335	0.359	0.404	0.435	0.451	0.444	0.436	0.388	0.362	0.325	0.312	
	taud	2.409	2.299	2.320	2.217	2.181	2.184	2.222	2.229	2.342	2.395	2.457	2.407	
	Ebn at Noon	269	274	281	274	268	263	264	262	268	262	258	254	
	Edh at Noon	25	33	36	43	46	46	44	43	35	30	24	23	
All-Sky Solar Radiation	RadAvg	486	775	1104	1469	1725	1921	1926	1691	1376	909	582	399	
	RadStd	51	103	98	123	148	130	108	62	109	73	79	39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page